CRICKETVERSE: A Decentralized Web3 Cricket Ecosystem

Course Project Proposal

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Academic Year: 2026

Project Metadata Overview

Item	Details
Project Title	CricketVerse
Project Type	B.Tech / Academic Web3 Prototype
Primary Domain	Blockchain, Sports Engineering, Gamification & Digital Assets
Smart Contract Stack	Solidity, ERC-20, ERC-721, ERC-1155
Execution Network	Polygon Amoy Testnet / EVM-compatible Testnet
Utility Token	CRIC (Prototype ERC-20 Token)

1. Executive Summary

Cricket represents one of the world's largest digital fan communities, boasting an estimated 2.5 billion global followers. While fans actively engage through fantasy leagues, digital collectibles, ticketing platforms, and social polling, these interactions currently occur across fragmented, centralized Web2 silos. Existing platforms suffer from operational black-boxing, closed scoring logic, custodial asset risks, and rampant secondary market ticket scalping.

As part of my Course Project, I propose CricketVerse, a unified, decentralized Web3 sports application designed to integrate fantasy gaming, dynamic Non-Fungible Token (NFT) collectibles, tamper-proof ticketing, and fan engagement into a single provably fair ecosystem. The platform utilizes smart contracts on an Ethereum-compatible test network to manage platform utility tokens (CRIC), digital player assets, and settlement pools, while maintaining high-throughput player telemetry off-chain in a traditional database.

The core architectural focus of my project is a hybrid state model: critical ownership records, escrow mechanisms, and marketplace transactions are secured on-chain, whereas high-volume telemetry, player stats, and analytics remain off-chain. This design mitigates unnecessary gas expenditure and optimizes performance, resulting in a realistic, scalable academic Web3 prototype.

2. Problem Statement & Motivation

2.1 Current Industry Limitations

Contemporary digital sports platforms force users into closed operational boundaries, presenting distinct structural flaws:

- **Custodial Ownership Risks:** In traditional fantasy apps, digital assets (cards, badges, passes) reside in centralized SQL databases. Users possess no true ownership and cannot export or monetize their assets outside the native app.
- **Opaque Execution & Logic:** Scoring mechanisms, payout distributions, and leaderboard calculations occur in closed backends, leaving users vulnerable to algorithmic manipulation or administrative delays.
- **Secondary Market Scalping & Fraud:** Ticket resale markets suffer from fake duplicate listings, excessive markup pricing, and lack of organizer royalty enforcement.
- **High Web3 Onboarding Friction:** Standard Web3 applications overload non-technical users with seed phrase management and upfront native gas token purchases (e.g., MATIC/ETH).

2.2 Core Research & Engineering Question

How can EVM-compatible smart contract infrastructure and dynamic token standards be architected to unify fantasy gaming, digital asset ownership, and event ticketing into a single, user-accessible sports platform without sacrificing transaction throughput or data privacy?

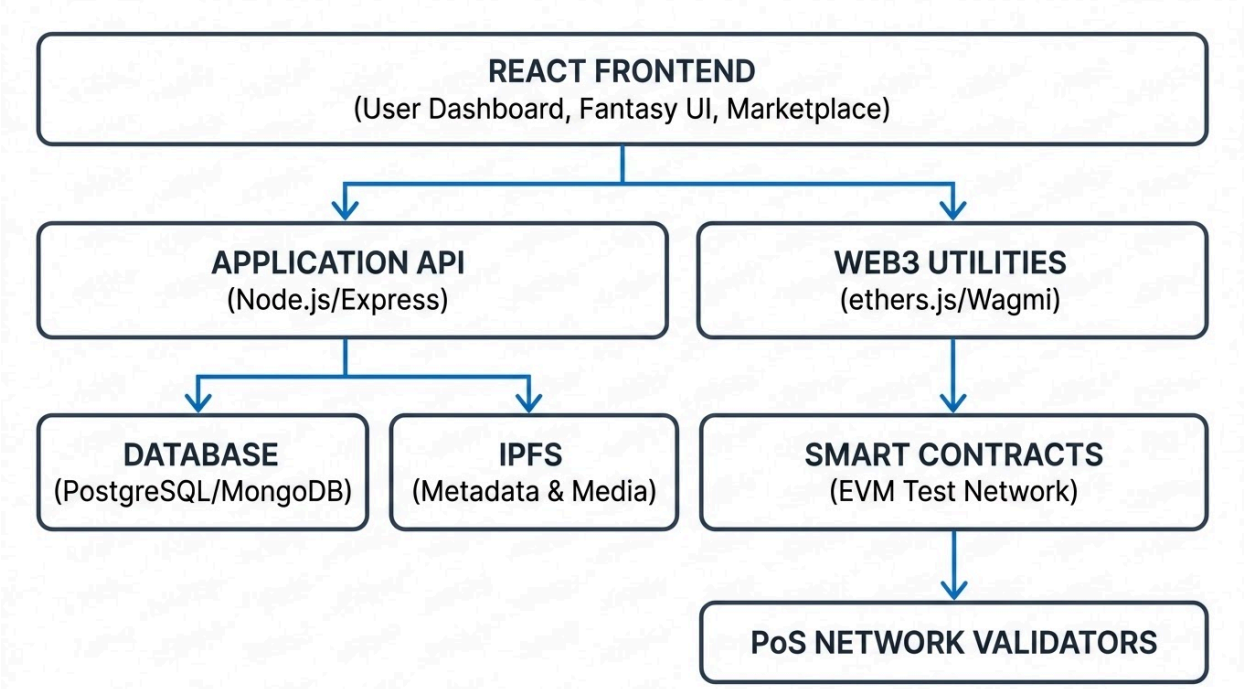
3. Project Objectives

To address the limitations outlined above, I have defined the following primary engineering objectives for CricketVerse:

1. **Architect a Unified Multi-Token Registry:** Deploy smart contracts (ERC-20, ERC-721, and ERC-1155) to manage platform utility tokens (CRIC), player card NFTs, venue tickets, and achievement badges.
2. **Develop an Oracle-Driven Fantasy Scoring Engine:** Integrate Chainlink Functions to fetch real-world ball-by-ball performance metrics, enabling deterministic, automated fantasy point distribution and non-custodial payouts.
3. **Implement Dynamic NFT Metadata Updates:** Program player cards whose performance multipliers and metadata automatically update based on real-world athletic milestones.
4. **Implement Anti-Scalping Ticket Logic:** Create ticket NFTs with enforced maximum secondary resale caps and automated venue royalty splits built directly into contract code.
5. **Formulate a Hybrid On/Off-Chain Architecture:** Optimize storage efficiency by maintaining transactional records on-chain while storing heavy analytics in PostgreSQL/MongoDB.

4. System Architecture & Methodology

CricketVerse leverages a high-performance hybrid architecture.
High-frequency operations (live scores, user search, leaderboards) run on conventional cloud infrastructure, while state settlement and ownership verification rely on EVM smart contracts.



4.1 Technology Stack Selection

Layer	Selected Technology	Technical Purpose
Presentation Layer	React.js, Tailwind CSS	Responsive dashboard, fantasy squad builder, marketplace UI
Backend API	Node.js / Express	Authentication, REST APIs, scoring calculation, IPFS pinning
Database	PostgreSQL / MongoDB	Player stats, match schedules, leaderboard indexes, caching
Blockchain	Polygon Amoy / EVM Testnet	State transitions, token issuance, marketplace execution

Layer	Selected Technology	Technical Purpose
Smart Contracts	Solidity (v0.8.x), OpenZeppelin	Standardized, audited token protocols & access control
Client Web3 Bridge	ethers.js / Wagmi	Contract reads/writes, signature verification, wallet connection
Decentralized Media	IPFS (Pinata)	Decentralized storage of card media and metadata JSON files

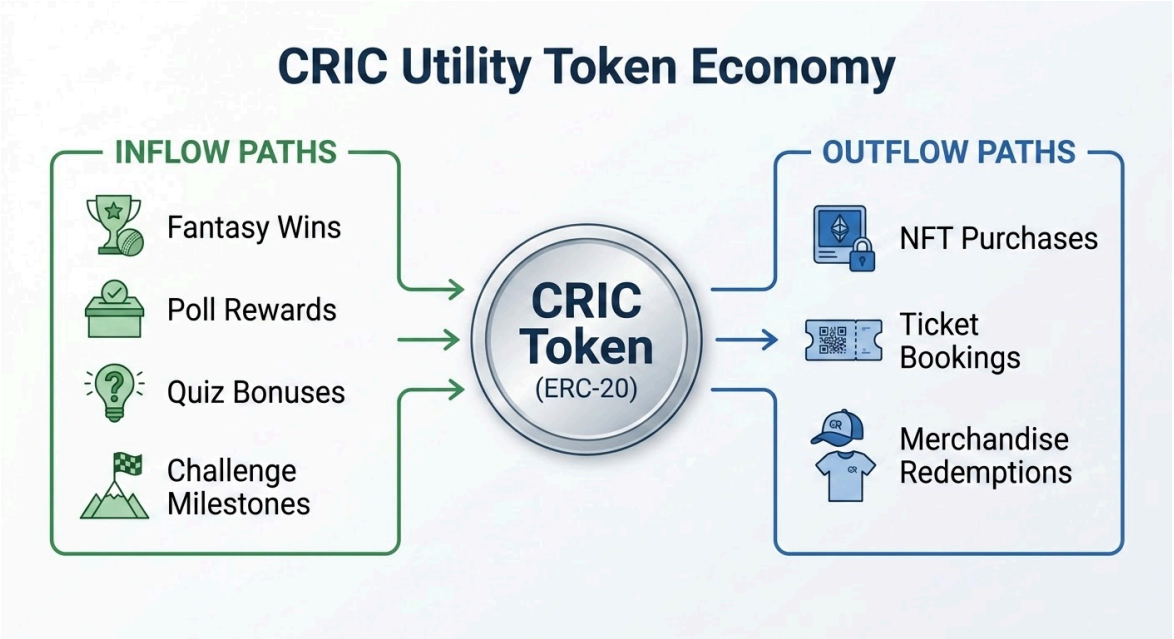
4.2 Nodes, Consensus, and Network Participants

- **User Wallet (MetaMask/AA Wallet):** Holds keys, signs state changes, and stores CRIC and NFT assets.
- **RPC Gateway / Node:** Relays signed payload data from the application layer to the test blockchain.
- **PoS Network Validators:** Produce and attest blocks on the underlying test network. *Note: As an Ethereum-compatible Proof-of-Stake chain, the network does not rely on application-level miners.*

5. Core Platform Modules

5.1 CRIC Token Economy

The CRIC token is an ERC-20 utility token serving as the internal accounting unit across the platform ecosystem.



- **Simulated On-Ramp:** For prototype testing, a mock fiat payment gateway allows users to test card and ticket purchases without real monetary risk.

5.2 Dynamic Player Card NFTs (dNFTs)

Player cards are issued under the ERC-1155 standard to support both fungible editions and unique cards.

- **Attributes & Metadata:** Each card encodes player role, batting/bowling stats, rarity tier, and total supply.
- **Rarity Distribution Matrix:**

Rarity Tier	Supply Cap	Multiplier	Special Attributes
Common	High / Unlimited	1.0x	Base player stats
Uncommon	5,000 units	1.1x	Slight performance boost
Rare	1,000 units	1.25x	Enhanced scoring potential

Rarity Tier	Supply Cap	Multiplier	Special Attributes
Epic	250 units	1.5x	Visual enhancements & boosted metrics
Legendary	25 units	2.0x	Highest rarity, special edition art

5.3 Peer-to-Peer Marketplace Engine

The non-custodial marketplace contract enables trustless trading of player cards using CRIC tokens.

- **Atomic Settlement:** The listing contract verifies asset ownership and spending approvals simultaneously, transferring funds and NFTs in a single atomic transaction.
- **Resale Royalty Enforcement:** Original card creators or drop pools receive a configurable royalty fee on secondary sales.

5.4 Fantasy Gaming & Score Calculations

Users construct lineups within a designated virtual salary cap. Points are generated deterministically from match statistics ingested via verified sports API feeds.

Match Performance Metric	Fantasy Points Awarded
Runs Scored	+1 Point per run
Boundary Bonus (4s)	+1 Extra Point
Sixer Bonus (6s)	+2 Extra Points
Wicket Taken	+20 Points
Catch Claimed	+8 Points

Match Performance Metric	Fantasy Points Awarded
Half-Century (50 runs)	+10 Milestone Bonus
Century (100 runs)	+25 Milestone Bonus

5.5 Anti-Scalping Ticket & Merchandise Module

Venue tickets and digital merchandise vouchers are issued as non-fungible tokens, allowing venue operators to maintain control over resale markets.

- **Resale Price Ceiling:** Smart contract parameters restrict secondary transfers above a defined markup threshold (e.g., maximum 15% above face value).
- **One-Time Redemption:** Venue gatekeepers check token ownership and signature validity; calling the redemption function permanently alters the ticket state to prevent double-entry.

6. On-Chain vs. Off-Chain Data Strategy

To maintain high throughput and minimize transaction costs, data is strictly categorized:

Data Element	Storage Location	Technical Justification
CRIC Balances & Transfers	On-Chain (ERC-20)	Requires auditability and trustless verification
NFT Ownership Records	On-Chain (ERC-7...	Immutable proof of digital ownership
Marketplace Settlement	On-Chain Smart ...	Eliminates counterparty risks during transactions

Data Element	Storage Location	Technical Justification
Ticket Redemption State	On-Chain Smart ...	Prevents duplicate usage and unauthorized scalping
Player Performance Stats	Off-Chain Database	High frequency data; cost-prohibitive on-chain
Card Artwork & Media	IPFS	Large binary file assets stored decentrally
Match Schedules & Leaderboards	Off-Chain Cache	Requires fast querying and low-latency rendering

7. Implementation Roadmap

The development schedule is structured across a comprehensive 16-week timeline:

Phase / Timeline	Development Focus & Deliverables
Weeks 01–02	SRS & System Architecture Specification
Weeks 03–04	Database Schema Design & Local Hardhat Network Setup
Weeks 05–06	CRIC (ERC-20) & PlayerCard (ERC-1155) Smart Contracts
Weeks 07–08	Marketplace Contracts & Frontend Web3 Integration
Weeks 09–10	Fantasy Scoring Engine & Off-Chain Indexer

Phase / Timeline	Development Focus & Deliverables
Weeks 11–12	Ticket NFTs, Gate Verification Portal & Fan Polls
Weeks 13–14	Security Auditing, Unit Testing & Testnet Deployment
Weeks 15–16	User Acceptance Testing, Evaluation & Final Project Documentation

8. Success Metrics & Security Plan

8.1 Verification and Testing Metrics

- **Smart Contract Test Coverage:** 100% pass rate across unit test suites built with Hardhat/Foundry.
- **State Transition Integrity:** Zero balance discrepancies or improper asset transfers during marketplace concurrency testing.
- **Scoring Engine Accuracy:** Deterministic point calculation validated against historical cricket match records.
- **Verification Speed:** Ticket validation and QR signature verification completed under two seconds at the gate portal.

8.2 Security Framework

- **OpenZeppelin Standards:** Use inherited, community-audited base contracts for token balances, access controls, and emergency pause logic.
- **Reentrancy Protection:** Enforce strict checks-effects-interactions patterns and apply `ReentrancyGuard` to all financial functions.
- **Access Control Controls:** Apply role-based privileges (`DEFAULT_ADMIN_ROLE`, `MINTER_ROLE`) to restrict administrative operations.

9. Conclusion

The CricketVerse prototype demonstrates how modern blockchain engineering can transform digital sports engagement. By combining fantasy gaming, dynamic player cards, controlled secondary marketplaces, and anti-scalping ticketing into a single hybrid platform, this project establishes a scalable foundation for trustless, transparent fan experiences.

References

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4. **Buterin, V.** (2014). *A Next-Generation Smart Contract and Decentralized Application Platform.* Ethereum Whitepaper.
5. **CricketVerse Initial Project Abstract & Proposal Documents** (Internal Academic Repository, CSE Department, 2026).